



3558 - The First Spectrum of the Coldest Halo Brown Dwarf

Cycle: 2, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSPEC				
	2	W1534-1034	NIRSpec Fixed Slit Spectroscopy	(1) W1534-1043
MIRI LRS				
	4	W1534-1034	MIRI Low Resolution Spectroscopy	(1) W1534-1043
	6	W1534-1034	MIRI Low Resolution Spectroscopy	(1) W1534-1043
MIRI Imaging				
	5	W1534-1034	MIRI Imaging	(1) W1534-1043

ABSTRACT

For nearly a decade, JWST GTO target WISE 0855 ($T_{\text{eff}} \sim 250$ K, $d \sim 2$ pc) has stood as the coldest known brown dwarf. Perplexingly, intense searches for WISE 0855 analogs have turned up no objects as cold or cooler to date. One emerging possibility is that the very coolest brown dwarfs — like giant (exo)planets — have diverse properties, and thus our past searches have been misguided. Because brown dwarfs cool over billions of years as they age, the coldest substellar objects may preferentially be the oldest, but until recently no halo brown dwarfs had been conclusively identified. WISEA 1534-1043 (a.k.a. The Accident) is a newly discovered brown dwarf with halo kinematics and observational properties completely unlike those of any other known (sub)stellar object, though strongly suggestive of low metallicity. At just 16 pc distant and with $T_{\text{eff}} < \sim 400\text{--}550$ K, WISEA 1534-1043 may be the first member of a vast, previously overlooked halo brown dwarf population. Our JWST NIRSpec/MIRI spectroscopy and MIRI photometry of WISEA 1534-1043 will (1) determine this unique object’s bolometric luminosity, temperature, and metallicity (2) complete WISEA 1534-1043’s kinematic profile (3) reveal chemistry at low metallicity via retrievals (4) serve as a touchstone data set steering future development of low-metallicity, low-temperature brown dwarf and exoplanet atmospheric/evolutionary models and (5) guide future searches for the coolest brown dwarfs. Incredibly faint at short wavelengths ($J = 24.5$ Vega), high-quality spectroscopy of WISEA 1534-1043 is infeasible with any facility other than JWST.

OBSERVING DESCRIPTION

In this JWST proposal we have targeted the coldest known halo brown dwarf candidate in order to obtain a full spectral energy distribution, calculate bolometric luminosity, complete a kinematic analysis with radial velocity, and investigate the complex chemistry resolved in JWST data. We propose to acquire NIRSpec G395H spectra along with low resolution MIRI and NIRSpec LRS spectra. The G395H grating covers the peak of the flux for our target where measurable discrepancies are expected compared to typical brown dwarfs. The NIRSpec and MIRI LRS data along with a few MIRI photometric data points will allow us to calculate a bolometric luminosity anchoring all observations with this fundamental observed parameter.

Proposal 3558 - Targets - The First Spectrum of the Coldest Halo Brown Dwarf

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	W1534-1043	RA: 15 34 29.2495 (233.6218729d) Dec: -10 43 18.39 (-10.72177d) Equinox: J2000	Proper Motion RA: -1253.1 mas/yr Proper Motion Dec: -2377 mas/yr Parallax: .0614" Epoch of Position: 2016.70	
Comments: Category=Star Description=[Brown dwarfs] Extended=NO					

Proposal 3558 - Observation 2 - The First Spectrum of the Coldest Halo Brown Dwarf

Observation	Proposal 3558, Observation 2: W1534-1034										Fri Aug 16 22:00:22 GMT 2024	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	W1534-1043	RA: 15 34 29.2495 (233.6218729d) Dec: -10 43 18.39 (-10.72177d) Equinox: J2000			Proper Motion RA: -1253.1 mas/yr Proper Motion Dec: -2377 mas/yr Parallax: .0614" Epoch of Position: 2016.70						
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	1 W1534-1043	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	146489	
Template	Slit					Subarray						
	S200A1					SUBS200A1						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	S200A1	NRSRAPID	28	18	1	NONE	3	54	2440.934	146489
	2	PRISM/CLEAR	S200A1	NRS	35	10	2	NONE	3	30	6590.954	146489

Proposal 3558 - Observation 4 - The First Spectrum of the Coldest Halo Brown Dwarf

Observation	Proposal 3558, Observation 4: W1534-1034								Fri Aug 16 22:00:22 GMT 2024	
	Diagnostic Status: Warning									
Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(1)	W1534-1043	RA: 15 34 29.2495 (233.6218729d)		Proper Motion RA: -1253.1 mas/yr					
			Dec: -10 43 18.39 (-10.72177d)		Proper Motion Dec: -2377 mas/yr					
			Equinox: J2000		Parallax: .0614"					
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	F1000W	FASTGRPAVG	8	1	1	88.801	146489	
Template	Subarray				Obtain Verification Image?					
	FULL				true					
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
	1	FASTR1	8	1	1	1	1	22.2		F1000W

Proposal 3558 - Observation 4 - The First Spectrum of the Coldest Halo Brown Dwarf

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	250	5	10	1	2	6959.8	146489

Proposal 3558 - Observation 6 - The First Spectrum of the Coldest Halo Brown Dwarf

Observation	Proposal 3558, Observation 6: W1534-1034									Fri Aug 16 22:00:22 GMT 2024
	Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	W1534-1043	RA: 15 34 29.2495 (233.6218729d) Dec: -10 43 18.39 (-10.72177d) Equinox: J2000			Proper Motion RA: -1253.1 mas/yr Proper Motion Dec: -2377 mas/yr Parallax: .0614" Epoch of Position: 2016.70				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	F1000W	FASTGRPAVG	8	1	1	88.801	146489	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
	1	FASTR1	8	1	1	1	1	22.2		F1000W

Proposal 3558 - Observation 6 - The First Spectrum of the Coldest Halo Brown Dwarf

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	250	5	10	1	2	6959.8	146489

Proposal 3558 - Observation 5 - The First Spectrum of the Coldest Halo Brown Dwarf

Observation	Proposal 3558, Observation 5: W1534-1034										Fri Aug 16 22:00:22 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	W1534-1043	RA: 15 34 29.2495 (233.6218729d)				Proper Motion RA: -1253.1 mas/yr				
			Dec: -10 43 18.39 (-10.72177d)				Proper Motion Dec: -2377 mas/yr				
			Equinox: J2000				Parallax: .0614"				
							Epoch of Position: 2016.70				
Template	Comments: Category=Star Description=[Brown dwarfs] Extended=NO										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	20	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F1500W	FASTR1	5	1	1	Dither 1	4	4	55.501	146489
	2	F1800W	FASTR1	8	1	1	Dither 1	4	4	88.801	146489
	3	F2100W	FASTR1	16	2	1	Dither 1	4	8	366.305	146489